Geometric Brownian Motion: Exact Terminal Sampler

`hasklib` mathematical companion

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Abstract

This note derives the result that `GBM::sample_terminal` relies on: applying Itô's lemma to $\ln X_t$ linearises the geometric Brownian motion SDE and yields an exact (not discretised) sampler for X_T given X_0. We also derive the first two moments, which are what `test_stochastic.cpp` checks against simulated output. This file assumes Itô's lemma as established in `ito_foundations.tex` (Theorem 1 there); it is not re-derived here. The derivation is classical; see Shreve (2004) for the standard textbook treatment.

1 The GBM SDE

Definition 1 (Geometric Brownian motion)**.** X_t follows geometric Brownian motion with drift $\mu \in \mathbb{R}$ and volatility $\sigma > 0$ if it solves

$$\mathrm{d}X_t = \mu X_t \,\mathrm{d}t + \sigma X_t \,\mathrm{d}W_t, \qquad X_0 > 0. \tag{1}$$

In the notation of `ito_foundations.tex` this is the choice $a(t,x) = \mu x$, $b(t,x) = \sigma x$ in the general SDE $\mathrm{d}X_t = a(t,X_t)\mathrm{d}t + b(t,X_t)\mathrm{d}W_t$.

Remark 1 (Correspondence with the code). This is exactly the pair of functions `GBM::drift` and `GBM::diffusion` return: `drift(t,x)` $= \mu x$, `diffusion(t,x)` $= \sigma x$. `EulerMaruyama` consumes these polymorphically through the `StochasticProcess` interface; the sampler derived below is the exact alternative to that discretisation, used by `GBM::sample_terminal` instead of stepping the Euler scheme forward.

Both a and b are linear in x and (for $x > 0$, which GBM never leaves since $X_t = X_0 \exp(\cdot) > 0$ once we derive the solution below) globally Lipschitz on any bounded interval away from 0, so the existence–uniqueness assumption of `ito_foundations.tex` applies and (1) has a unique strong solution.

2 Linearising via Itô's lemma

The SDE (1) has no constant-coefficient closed form as written, because both coefficients scale with X_t. The standard fix is to apply Itô's lemma to $f(t,x) = \ln x$: this removes the x-dependence from both coefficients and turns (1) into an SDE with constant drift and diffusion, which integrates directly.

Theorem 1 (Exact terminal law of GBM)**.** *Let X_t solve* (1)*. Then for any $T > 0$,*

$$X_T = X_0 \exp\!\left(\left(\mu - \tfrac{1}{2}\sigma^2\right)T + \sigma W_T\right), \tag{2}$$

where $W_T \sim \mathcal{N}(0,T)$.

Proof. Apply Itô's lemma (Theorem 1 of `ito_foundations.tex`) to $\varphi(t,x) = \ln x$, so $\frac{\partial\varphi}{\partial t} = 0$, $\frac{\partial\varphi}{\partial x} = \frac{1}{x}$, $\frac{\partial^2\varphi}{\partial x^2} = -\frac{1}{x^2}$. With $a(t, X_t) = \mu X_t$ and $b(t, X_t) = \sigma X_t$,

$$\mathrm{d}(\ln X_t) = \Big(0 + \mu X_t \cdot \frac{1}{X_t} + \tfrac{1}{2}\,\sigma^2 X_t^2 \cdot \Big(-\frac{1}{X_t^2}\Big)\Big)\mathrm{d}t + \sigma X_t \cdot \frac{1}{X_t}\,\mathrm{d}W_t.$$

The X_t factors cancel completely, leaving

$$\mathrm{d}(\ln X_t) = \Big(\mu - \tfrac{1}{2}\sigma^2\Big)\mathrm{d}t + \sigma\,\mathrm{d}W_t. \tag{3}$$

The right-hand side no longer depends on X_t: it is Brownian motion with constant drift $\mu - \frac{1}{2}\sigma^2$ and constant volatility σ, so (3) integrates directly from 0 to T:

$$\ln X_T - \ln X_0 = \Big(\mu - \tfrac{1}{2}\sigma^2\Big)T + \sigma\,(W_T - W_0) = \Big(\mu - \tfrac{1}{2}\sigma^2\Big)T + \sigma W_T,$$

using $W_0 = 0$. Exponentiating gives (2). Since $W_T \sim \mathcal{N}(0, T)$ by definition of Brownian motion, this is a complete description of the law of X_T given X_0. □

Remark 2. Equation (2) is sampled with a single normal draw: generate $Z \sim \mathcal{N}(0,1)$, set $W_T = \sqrt{T}\,Z$, and evaluate the right-hand side. No time-stepping is needed, which is what makes it an exact sampler rather than a discretisation, and it is exactly what `GBM::sample_terminal` implements using `NormalRng::draw()`.

Remark 3 (Where the $-\frac{1}{2}\sigma^2$ comes from). The drift that appears in (2) is not μ, the drift of X_t itself, but $\mu - \frac{1}{2}\sigma^2$. That correction is exactly the $\frac{1}{2}b^2\frac{\partial^2\varphi}{\partial x^2}$ term of Itô's lemma, which is the term with no analogue in ordinary calculus, discussed at length in `ito_foundations.tex`. If one (incorrectly) applied the ordinary chain rule to $\ln X_t$, the correction would be missing and the resulting sampler would be biased upward in expectation, as proposition 1 below makes precise.

3 Moments

The moment-matching test in `test_stochastic.cpp` checks simulated terminal samples against the following closed-form mean and variance.

Lemma 1 (Moment generating function of a Gaussian).
If $Y \sim \mathcal{N}(m, v)$ *then* $\mathbb{E}[e^Y] = \exp\big(m + \frac{1}{2}v\big)$.

Proof.
Write $Y = m + \sqrt{v}\,Z$ with $Z \sim \mathcal{N}(0,1)$. Complete the square in the Gaussian integral:

$$\mathbb{E}[e^Y] = e^m\,\mathbb{E}\big[e^{\sqrt{v}\,Z}\big] = e^m \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}\,e^{\sqrt{v}\,z - \frac{1}{2}z^2}\,\mathrm{d}z = e^m\,e^{\frac{1}{2}v} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}\,e^{-\frac{1}{2}(z-\sqrt{v})^2}\,\mathrm{d}z = e^{m+\frac{1}{2}v},$$

the last integral being 1 since it is a $\mathcal{N}(\sqrt{v}, 1)$ density integrated over $\mathbb{R}$. □

Proposition 1 (Mean of GBM). $\mathbb{E}[X_T] = X_0\,e^{\mu T}$.

Proof.
By theorem 1, $X_T = X_0\,e^Y$ with $Y = (\mu - \frac{1}{2}\sigma^2)T + \sigma W_T$. Since $W_T \sim \mathcal{N}(0,T)$, $Y \sim \mathcal{N}\big((\mu - \frac{1}{2}\sigma^2)T,\ \sigma^2 T\big)$. Applying lemma 1 with $m = (\mu - \frac{1}{2}\sigma^2)T$ and $v = \sigma^2 T$,

$$\mathbb{E}[X_T] = X_0\,\mathbb{E}[e^Y] = X_0 \exp\Big((\mu - \tfrac{1}{2}\sigma^2)T + \tfrac{1}{2}\sigma^2 T\Big) = X_0\,e^{\mu T}.$$

The $-\frac{1}{2}\sigma^2 T$ Itô correction and the $+\frac{1}{2}\sigma^2 T$ from the moment generating function cancel exactly, leaving the mean growing at the "natural" rate μ despite the correction appearing inside every individual path. □

Proposition 2 (Variance of GBM). $\mathrm{Var}(X_T) = X_0^2\, e^{2\mu T}\big(e^{\sigma^2 T} - 1\big)$.

Proof. $\mathrm{Var}(X_T) = \mathbb{E}[X_T^2] - \big(\mathbb{E}[X_T]\big)^2$. For the second moment, write $X_T^2 = X_0^2\, e^{2Y}$ with Y as above, so $2Y \sim \mathcal{N}\big(2(\mu - \frac{1}{2}\sigma^2)T,\ 4\sigma^2 T\big)$. By lemma 1 with $m = 2(\mu - \frac{1}{2}\sigma^2)T$ and $v = 4\sigma^2 T$,

$$\mathbb{E}[X_T^2] = X_0^2 \exp\Big(2\big(\mu - \tfrac{1}{2}\sigma^2\big)T + 2\sigma^2 T\Big) = X_0^2\, e^{(2\mu+\sigma^2)T}.$$

Subtracting $\big(\mathbb{E}[X_T]\big)^2 = X_0^2\, e^{2\mu T}$ from proposition 1 and factoring,

$$\mathrm{Var}(X_T) = X_0^2\, e^{(2\mu+\sigma^2)T} - X_0^2\, e^{2\mu T} = X_0^2\, e^{2\mu T}\big(e^{\sigma^2 T} - 1\big). \qquad \square$$

Remark 4 (What the test checks). `test_stochastic.cpp` draws a large number of terminal samples from `GBM::sample_terminal`, computes their sample mean and sample variance, and checks these against propositions 1 and 2 to within Monte Carlo tolerance. This is a check on the sampler, not on (1) itself — it would catch, for instance, a sign error in the $-\frac{1}{2}\sigma^2$ correction, since that error changes the mean but leaves the qualitative shape of the distribution intact.

References

Shreve, S. E. (2004), *Stochastic Calculus for Finance II: Continuous-Time Models*, Springer Finance, Springer, New York.